Clustering and Topic Discovery

POS6933: Computational Social Science

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Overview

- Housekeeping
- Class 6 Problem Set Review
- Discriminating Words
- Clustering
- Topic Models (LDA & STM)

Housekeeping

- Final Paper Drafts Due 3/29 (*Optional*)
- Presentation & Peer Assessment Schedule Coming 3/23
- Will canvas ideas for 4/6 class on 3/23

Class 6 Problem Set Review

- Notes...

Discriminating Words

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- **Discrimination Words**: The frequency and distribution of words carry meaningful information about latent categories – i.e., words that appear more frequently in some categories than others are informative signals about that category
- Ex: *Healthcare Provider* vs. *Abortionist* – Terms used in the same debate but relay competing sentiments/connotations/frames

Discriminating Words – Competing Terms Example (D v. R)

Democrats	Republicans
Climate	Freedom
Healthcare	Tax
Immigration	Border
Justice	Security
Welfare	Economy

- This is <u>not</u> to say these words invariably represent parties (e.g., useful in vocabulary for dictionary/class assignment)

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- This is <u>not</u> to say these words invariably represent parties (e.g., useful in vocabulary for dictionary/class assignment)
- Rather – Words are useful to help us discriminate (distinguish) between them.
- Ex: Democrats can use *Border*, too (similar or different frames) – but more likely to appear as term used in by Republicans.

Mutual Information

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- MI $= 0 \sim$ Information between two variables (ex: a label and a word) are completely independent (bad discriminator)
- MI $= 1 \sim$ Knowing about one variable completely informs us of the other (good discriminator).

Calculating Mutual Information

- k = Categories
- j = Word
- π_k = Proportion of Documents for k
- π_{-k} = Probability that Doc Doesn't Belong in k

Calculating Mutual Information (Cont.)

$$H(k) = -\pi_k \log_2 \pi_k - \pi_{-k} \log_2 \pi_{-k},$$

Unconditional Certainty: *How likely are to assume any random document belongs to a particular category without even seeing a word*

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$$H(k|j) = -\sum_{k,-k} \sum_{j,-j} \pi_{k,j} \log_2 \pi_{k,j}$$

Conditional Certainty

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Conditional Certainty

$$\text{MI}_{kj} = H(k) - H(k|j)$$

Mutual Information = Unconditional Certainty - Conditional Certainty

Calculating Mutual Information (Example)

Doc ID	Category (k)	Words (j)
1	Red	Apple, Tree
2	Red	Apple, Juice
3	Red	Tree, Green
4	Red	Apple, Green
5	Blue	Blue, Ocean
6	Blue	Ocean, Water
7	Blue	Blue, Water
8	Blue	Ocean, Blue
9	Red	Apple, Tree
10	Blue	Ocean, Tree

Calculating Mutual Information (Example - Cont.)

Doc ID	Category (k)	Words (j)
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2	Red	Apple, Juice
3	Red	Tree, Green
4	Red	Apple, Green
5	Blue	Blue, Ocean
6	Blue	Ocean, Water
7	Blue	Blue, Water
8	Blue	Ocean, Blue
9	Red	Apple, Tree
10	Blue	Ocean, Tree

Calculate Unconditional Certainty (π_k)

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Calculate Unconditional Certainty (π_k)

$$\pi_{Red} = 5/10 = 0.5$$

$$\pi_{Blue} = 5/10 = 0.5$$

Calculating Mutual Information (Example - Cont.)

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$$H(K) = -\pi_{Red} \log_2 \pi_{Red} - \pi_{Blue} \log_2 \pi_{Blue}$$

$$H(K) = -0.5 \log_2 0.5 - 0.5 \log_2 0.5$$

$$H(K) = 1$$

Calculating Mutual Information (Example - Cont.)

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Calculate Conditional Certainty For Tree (π_k)

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Calculate Conditional Certainty For Tree (π_k)

$$P(Red|Tree) = 3/4 = 0.75$$

$$P(Blue|Tree) = 1/4 = 0.25$$

$$H(K) = -\pi_{Red}\log_2\pi_{Red} - \pi_{Blue}\log_2\pi_{Blue}$$

Calculating Mutual Information (Example - Cont.)

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1	Red	Apple, Tree
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Calculate Conditional Certainty For Tree (π_k)

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$$H(K) = -\pi_{Red}\log_2\pi_{Red} - \pi_{Blue}\log_2\pi_{Blue}$$

$$H(K|Tree) = -(0.75\log_2 0.75 + 0.25\log_2 0.25)$$
$$= 0.811$$

$$MI(k, Tree) = H(k) - H(k|Tree)$$
$$= 1 - 0.811 \approx 0.189$$

Calculating Mutual Information (Example - Cont.)

- **What if we tried with *Green*?**

Calculating Mutual Information (Example - Cont.)

- **What if we tried with *Green*?**

  $P(Red|Green) = 2/2 = 1$

  $P(Blue|Green) = 0/2 = 0$

- *Tree* is the only word we can use to discriminate – It's the only term that appears in both categories!

Fightin' Words (Monroe, Coloresi, and Quinn 2008)

- Alternative strategy to identify words that discriminate b/w categories using regularized log-odds ratios – α_j = Dirichlet prior & $W_k^* \sim$ Multinomial(n_k, μ_k)

$$\hat{\mu}_{jk} = \frac{W_{jk}^* + \alpha_j}{n_k + \sum_{j=1}^{j} \alpha_j}$$

- MI identifies words that are most informative about category membership overall – Fightin' Words stimates which words are disproportionately distinctive between groups by accounting for baseline frequencies and sampling variability.
- Estimate corresponds to expected value of posterior distribution $p(\mu|W)$ – $\hat{\mu}_{jk}$ is smoothed probability of word j in category k, allowing for small prior (e.g., Dirichlet).

$$\text{Log Odds Ratio}_{kj} = \log(\frac{\mu_{kj}}{1-\mu_{kj}}) - \log(\frac{\mu - kj}{1-\mu_{-kj}})$$

Fightin' Words (Monroe, Coloresi, and Quinn 2008)

Word	Red	Blue	Total
Apple	4	0	4
Tree	3	1	4
Green	2	0	2
Juice	1	0	1
Blue	0	3	3
Ocean	0	4	4
Water	0	2	2

- Assuming $\alpha_j = 0.5$ for all – Let's

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Blue	0	3	3
Ocean	0	4	4
Water	0	2	2

- Assuming $\alpha_j = 0.5$ for all – Let's
- $\hat{\mu}(Apple|Red) = \frac{n_{(Apple,Red)}+\alpha_{Apple}}{n_{Red}+\sum_j \alpha_j}$
- $\hat{\mu}(Apple|Red) = \frac{4+0.5}{10+7\times 0.5} = 0.33$

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- **Let's solve for** $\hat{\mu}(Apple|Blue)$

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- $\hat{\mu}(Apple|Red) = \frac{n_{(Apple,Red)}+\alpha_{Apple}}{n_{Red}+\sum_j \alpha_j}$
- $\hat{\mu}(Apple|Red) = \frac{4+0.5}{10+7\times 0.5} = 0.33$
- **Let's solve for** $\hat{\mu}(Apple|Blue)$
- $\hat{\mu}(Apple|Blue) = \frac{0+0.05}{9+7\times 0.5} = 0.04$

Fightin' Words (Cont.)

- Now that we have $\hat{\mu}(\text{Apple}|\text{Red}/\text{Blue})$, the log odds ratio that **Apple** is a predictor of **Red** is:

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- Now that we have $\hat{\mu}(\text{Apple}|\text{Red}/\text{Blue})$, the log odds ratio that **Apple** is a predictor of **Red** is:
- $\delta Red = \log(\frac{0.33}{1-0.33}) \rightarrow \log(0.492) \approx -0.70$
- $\delta Blue = \log(\frac{0.04}{1-0.04}) \rightarrow \log(0.041) \approx -3.19$
- $\delta_{kj} = -0.70 - (-3.19) \approx 2.49$

Fightin' Words (Cont.)

- Now that we have $\hat{\mu}(\text{Apple}|\text{Red}/\text{Blue})$, the log odds ratio that **Apple** is a predictor of **Red** is:
- $\delta Red = \log(\frac{0.33}{1-0.33}) \rightarrow \log(0.492) \approx -0.70$
- $\delta Blue = \log(\frac{0.04}{1-0.04}) \rightarrow \log(0.041) \approx -3.19$
- $\delta_{kj} = -0.70 - (-3.19) \approx 2.49$
- $exp(2.49) \approx 12$x – Meaning that the odds **Apple** appears in **Red** documents is approx. 12 times larger than **Blue** – because it's regularized we know its not just the result of an imabalance in observations for Red vs. Blue